Project Idea Pitch

Cyber Guardians

Software, Cloud and Cybersecurity Team

Team Members

Name	Role
Puleng Machete	Developer
Moribishane Ramonyai	Developer
Katlego Mojapelo	Developer
Kgomoewana Lethabo	Developer
Kgotso Marapyana	Cloud Engineer
Tshepo Letsoalo	Security Lead
Tshepang Chuene	Security Lead

Problem Overview

Educational institutions still rely on paper based processes for attendance and WIL hour tracking, and they struggle with IT support request management

These methods are vulnerable to lost records, inaccurate reporting, signature fraud and verification challenges for remote students.

Idea 1: WIL Tracker

Tracks the 320 WIL hours using GPS, photos and mentor approval.

Idea 1 Problem

Paper logbooks can be stolen or get lost

Excel records can be modified

Mentor signatures can be forged

Progress of 320 hours can be difficult to monitor.

Remote attendance verification is challenging.

Idea 1 Users

Students: Record hours and monitor progress.

Mentors: Verify and approve hours.

Coordinators: Monitor overall WIL completion.

Administrators: Manage users and system configuration.

Idea 1 Features

Clock in/out: feature that allows students to digitally record when they start and finish work.

GPS capture: this feature confirms where the student was when attendance was recorded.

Photo evidence: this feature allows students to upload proof of attendance or completed work.

Dashboard: this dashboard feature gives students a real-time view of their WIL progress.

Azure Architecture

App Service: Hosts the web application.

Azure SQL: Stores attendance and WIL records.

Blob Storage: Stores evidence photos.

Entra ID: Manages authentication and RBAC.

Key Vault: Stores secrets and credentials securely.

Security Controls

Role Based Access Control

Multi-Factor Authentication

Encryption

Audit logs

POPIA compliance

Policies

STRIDE Threat Model

Spoofing

Tampering

Repudiation

Information Disclosure

DoS

Elevation of Privilege.

OWASP ASPVS Security Considerations

Authentication

Access control

Session management

Logging controls.

Idea 2: IT Support Ticket System

Users report IT issues and track resolution through a ticket workflow.

Idea 2 Problem

Verbal reports can be forgotten.

No formal tracking system exists.

Users cannot monitor issue progress.

IT teams struggle to manage workload.

Idea 2 Users

Students: Submit support requests.

Staff: Report technical issues.

Technicians: Resolve support requests.

IT Administrators: Monitor system performance.

Idea 2 Features

Submit tickets

Dashboards

Secure Login

Assign Technician

Status Updates

Ticket History

Notifications

Idea 2
Cloud & Security

Cloud

Azure App Service

Azure SQL Database

Backup Services

Security

MFA

RBAC

Audit Logging

Encryption

Account Protection

Comparison Matrix

Criteria	WIL Tracker	IT Ticket System
CTU Relevance	High	Medium
Cybersecurity Scope	High	High
Cloud Computing Scope	High	High
Business Value	High	High

Recommend Idea: Secure Student WIL Hours Tracker.

Reasons

Supports CTU's 320-hour WIL requirement.

Solves a real student problem.

Strong cybersecurity focus.

Strong cloud implementation.

Demonstrates Software, Cloud and Cybersecurity collaboration.

Conclusion

Project Ideas Presented

Idea 1: Secure Student WIL Hours Tracker

Idea 2: Secure IT Support Ticket System

Team Recommendation:

Secure Student WIL Hours Tracker